Connected Denotation and Contextual Representation:
Prābhākara Sentence Semantics as Empirical Hypothesis in Transformer Architectures

Suresh Kamath Bola
Cornell University
sk2864@cornell.edu
Draft: March 2026

Abstract

For roughly seven centuries, the Prābhākara and Bhāṭṭa schools of Mīmāṃsā debated a foundational question about language: does sentence meaning arise from independently meaningful words that then combine (*abhihitānvaya*), or do words mean only as already embedded in sentential context (*anvitābhidhāna*)? This paper argues that the debate identifies structural properties of language processing that map productively onto transformer architectures. The paper develops a precise mapping between the Mīmāṃsā triad of interpretive conditions (*ākāṅkṣā, yogyatā, sannidhi*) and functionally distinct components of the transformer, distinguishing three senses of compositionality and showing that the Prābhākara critique targets the central sense in which transformers depart from classical computational pipelines. Four experimental studies on BERT, GPT-2, and LLaMA 3.1 8B test predictions derived from this mapping. The paper finds that obligatory syntactic dependencies are tracked by 65% of BERT's attention heads yet prove causally redundant when individually blocked, consistent with constitutive rather than supplementary encoding. Semantic fitness is computed as a graded, distributed process concentrated in MLP layers, refining the classical binary conception. Contiguity operates both as a hard architectural boundary and as a soft exponential gradient with model-specific decay radii. Polysemy representations in BERT cross from identity-dominated to sense-dominated organization at layer 6 (bootstrap 95% CI [5, 8]), disconfirming a specific prediction of *abhihitānvaya*. The results demonstrate that placing a centuries-old debate in Indian philosophy of language in dialogue with mechanistic interpretability identifies structural distinctions precise enough to generate and evaluate testable hypotheses about neural language processing.

Key Sanskrit Terms

The following terms from the Mīmāṃsā tradition appear frequently. Mīmāṃsā is one of the six orthodox (*āstika*) schools of Indian philosophy, primarily concerned with the exegesis of Vedic texts. Its sustained engagement with scriptural interpretation produced some of the most detailed pre-modern theories of meaning in any tradition. Definitions are given here for convenience and elaborated in relevant sections.

Abhidhā: The denotative or referential function of a word. The capacity of a word to point to its referent.

Abhihitānvaya: Literally "denotation, then connection." The Bhāṭṭa Mīmāṃsā theory that words first denote independently and then combine to form sentence meaning.

Ākāṅkṣā: Mutual syntactic and semantic expectancy between words. A verb expects a subject; a noun in accusative case expects a transitive verb. One of three accepted conditions for sentential unity.

Anvitābhidhāna: Literally "connected denotation." The Prābhākara theory that words denote relationally, always as already connected to other words in the sentence.

Sannidhi: Contiguity or proximity. Words must be encountered close together in time or text for sentential meaning to arise. A word spoken today and a word spoken tomorrow cannot form a sentence.

Sphoṭa: Literally "that which bursts forth." In Bhartṛhari's philosophy of language, the indivisible meaning-bearing unity of a sentence, which is revealed (not constructed) through the sequence of phonemes.

Vākya: Sentence. The primary unit of linguistic communication in most Indian semantic theories.

Yogyatā: Semantic fitness or compatibility. The requirement that combined word meanings be non-absurd. The standard example of failure is vahninā siñcati ("irrigates with fire"), which is grammatically well-formed but semantically incoherent.

1. Introduction

Does the meaning of a sentence arise from the combination of independently meaningful parts, or do words mean only as already embedded in sentential context? This question is old. It was debated for roughly seven centuries within the Mīmāṃsā tradition of Indian philosophy, where it produced what McCrea (2000) called the most intricate and sophisticated system of language interpretation in pre-modern South Asia. It was central to the Frege-Wittgenstein trajectory in Western analytic philosophy (Lenci, 2008). And it is, in a precise technical sense, the question raised by the transformer architecture (Vaswani et al., 2017) that now dominates computational language processing.

For decades, the dominant computational approach to language followed a compositional pipeline: represent words as discrete symbols with fixed meanings, then combine them according to syntactic rules. This approach mirrors the compositional semantics developed in the tradition of Frege (1892) and Montague (1973). The transformer introduced a different processing structure. In a transformer, each token's representation, after the first layer of self-attention, is computed as a function of its relationships to every other token in the input sequence. There is no explicit "first represent, then compose" step.

This architectural feature has a philosophical parallel. Within Mīmāṃsā, the Prābhākara school developed a theory of sentence meaning (*anvitābhidhāna*) that rejects the sequential model of denotation followed by combination. On this view, the denotative function of a word is always relational: a word's meaning is not fixed in isolation and then composed, but is constituted in connection with other words in the sentence. The opposing view, *abhihitānvaya*, associated with the Bhāṭṭa school, holds that words first denote independently and then combine (McCrea, 2020, pp. 353-354). This debate, sustained over roughly seven centuries, was conducted with a shared set of conditions (*ākāṅkṣā*, *yogyatā*, *sannidhi*) accepted by both sides (Raja, 1963, pp. 149–187).

This paper examines whether these Mīmāṃsā conditions can be usefully mapped onto components of the transformer attention mechanism, and whether placing the two frameworks in dialogue yields not merely a re-description but testable predictions, philosophical arguments with explicit premises, and a vocabulary for describing what happens inside a transformer when it processes language. The central thesis is threefold. First, the three conditions (*ākāṅkṣā*, *yogyatā*, *sannidhi*) can be treated as procedural conditions on interpretation. Second, transformer behavior can serve as a testbed for

whether such conditions can be realized without explicit symbolic composition. Third, the experiments reported here provide evidence about which aspects of the classical theory are robust, which are distributed, and which are architecturally variable.

**** Related Interdisciplinary Work***

Several previous studies have compared language models to philosophical theories of meaning. Distributional semantics has been linked to Wittgenstein's use-theoretic account and to Firth's (1957, p. 11) maxim that "you shall know a word by the company it keeps" (Lenci, 2008, p. 6). Brandom's inferentialism has been proposed as a framework for language model behavior (Arai & Tsugawa, 2024). Dynamic semantics in the tradition of Heim (1982) and Groenendijk and Stokhof (1991) has been invoked for modeling contextual update across sequences (Zhu et al., 2025). Hermeneutic approaches drawing on Gadamer have likewise been applied to generative AI interpretation (Hornby, 2025). This paper differs from all of these in that it draws on a non-Western philosophical tradition, and specifically on an internal debate within that tradition between two rival theories that share a common set of conditions, rather than mapping a single framework onto language model behavior from the outside.

2. Three Senses of Compositionality

Discussions of compositionality in both philosophy and machine learning frequently conflate three distinct claims. Distinguishing them is a prerequisite for the argument that follows.

Semantic compositionality is the principle that the meaning of a complex expression is a function of the meanings of its parts and their syntactic mode of combination. Formally: for a complex expression e with constituents $e_1 \ldots e_n$ and structure S, the meaning $m(e) = f(m(e_1), \ldots, m(e_n), S)$. This is a constraint on the meaning function itself. See Szabó (2020) for a careful survey.

Procedural compositionality is a claim about the cognitive or computational process: that there exists a stage at which the meanings of parts are represented independently of their combination. A system could satisfy the semantic principle (its outputs could be described as a function of part-meanings) without there being any computational stage at which those part-meanings exist in isolation. The distinction is analogous to the difference between a function being decomposable in principle and a computation actually proceeding through the decomposed steps. Fodor and Pylyshyn (1988),

arguing from systematicity, treated this as a cognitive requirement; Smolensky (1990) and Smolensky and Legendre (2006) developed alternatives.

Systematic generalization is the empirical capacity to process novel combinations of familiar components correctly. Lake and Baroni (2018) argued that neural networks fail at this; subsequent work has partially contested the claim (Csordás et al., 2021; Ontanon et al., 2022).

The Prābhākara critique targets procedural compositionality. The Prābhākaras do not deny that sentence meanings can be analytically decomposed into word-level contributions after the fact. Their claim is that no cognitive act of grasping isolated word meanings precedes the act of grasping connected sentential meaning. As will be shown in Section 4, the transformer architecture similarly lacks an explicit sequential "first-represent-independently, then-compose" pipeline. Whether the absence of such a pipeline in the transformer constitutes a philosophically interesting parallel, or merely a superficial resemblance, is the central question of this paper. The experiments in Section 6 are designed to move the answer from speculation to evidence.

3. The Mīmāṃsā Debate on Sentence Meaning

**** Philosophical Context***

Mīmāṃsā takes as its primary task the interpretation of Vedic injunctions. The philosophical demands of scriptural exegesis forced its practitioners to develop detailed theories of how sentences mean. The central question was practical: given a Vedic sentence such as *svargakāmo jyotiṣṭomena yajeta* ("one who desires heaven should perform sacrifice with the *jyotiṣṭoma*"), by what process do the individual word meanings combine into a unified injunction (McCrea, 2000, p. 435)?

Two sub-schools formed around this question. The Bhāṭṭa school, following Kumārila Bhaṭṭa (c. 7th century CE), developed *abhihitānvaya*. The Prābhākara school, following Prabhākara Mishra (c. 7th century CE) and elaborated by Śālikanātha in the *Prakaraṇapañcikā* (c. 9th century CE) developed *anvitābhidhāna*. Both schools accept that words contribute to sentence meaning, but disagree on the order and nature of the cognitive operations involved (Matilal & Sen, 1988, p. 85).

**** Abhihitānvaya: Denotation, Then Connection***

Kumārila's position, developed in the Vākyapariccheda section of his *Ślōkavārttika*, holds that sentence meaning arises through two stages. In the first stage, each word activates its denotation

(*abhidhā*) independently. The word *gauh* ("cow") denotes the universal cowness (*gotva*) and, through that universal, a particular cow. This denotation holds regardless of sentential environment. In the second stage, a separate cognitive operation combines these independent denotations into a unified sentential meaning. The combination is governed by the three conditions of *ākāṅkṣā*, *yogyatā*, and *sannidhi* (Jha, 1942, pp. 176-221).

This is compositional in the procedural sense defined above: an identifiable stage of independent representation precedes an identifiable stage of combination.

**** Anvitābhidhāna: Connected Denotation***

Śālikanātha's *Prakaraṇapañcikā* rejects this two-stage model. In the Vākyapariccheda section, the argument runs roughly as follows (here reconstructed from the primary text and from Matilal's 1990 discussion, pp. 112-115):

Premise 1: If words first denote independently, then there should be a phenomenologically identifiable moment in which a hearer grasps an isolated word meaning before grasping its sentential role.

Premise 2: Introspection on the experience of sentence comprehension reveals no such moment. Understanding is, from the first moment of recognition, understanding-in-context.

Premise 3: A theory of meaning should be consistent with the phenomenology of comprehension.

Conclusion: Words do not first denote independently. Their denotative function (*abhidhā*) already includes syntactic and semantic connection to other words in the sentence.

A note on the evidential role of introspection: Premise 2 is the most contestable step. Contemporary philosophy of mind offers reasons for skepticism about introspective access to cognitive processes (Schwitzgebel, 2008). The Prābhākara argument would be strengthened by non-introspective evidence for the same conclusion. If both introspection and computational modeling point toward the same processing structure, the case for that structure becomes more robust, even if neither line of evidence is conclusive on its own. The experiments in Section 6 provide such non-introspective evidence.

The practical upshot of *anvitābhidhāna* is illustrated by a standard example. When the sentence *gām ānaya* ("bring the cow") is heard, the Prābhākara holds that the hearer does not first grasp "cow" and "bring" as independent meanings and then assemble them. Rather, *gām* (cow) denotes a-cow-as-to-

be-brought and *ānaya* (bring) denotes bringing-as-directed-at-a-cow. Denotation is relational (Raja, 1963, pp. 191-203).

Prabhākara himself addressed the case of words heard in isolation. If a single word is shouted across a marketplace, does the hearer grasp a context-free denotation? His answer, preserved in later commentarial literature (Deshpande, 1998), is that even then the hearer automatically supplies a minimal sentential context from memory and linguistic habit. Pure isolation is, on this view, a grammarian's abstraction.

It should be noted that *anvitābhidhāna* is stronger than what is found in contemporary contextualist philosophy. Recanati (2004) holds that lexical items are semantically underdeterminate and require contextual enrichment, but still assumes that something like a schematic lexical meaning exists prior to enrichment. The Prābhākara view denies even this: the denotative act is constitutively relational. No pre-contextual core exists to be enriched.

** The Three Conditions*

Both schools accept three conditions for a sequence of words to constitute a meaningful sentence. Where the schools differ is in the *location* of these conditions within the process of comprehension.

For the Bhāṭṭa school, the three conditions govern the second stage: the combination of independently grasped meanings. Words first denote; then *ākāṅkṣā*, *yogyatā*, and *sannidhi* regulate how those denotations are assembled.

For the Prābhākara school, the three conditions operate within denotation itself. The word *gām* denotes in a way that already reflects its *ākāṅkṣā*-relation to the verb, its *yogyatā*-compatibility with what is predicated of it, and its *sannidhi*-proximity to its neighbors. This difference in where the conditions are located is the crux of the debate.

4. Self-Attention and the Context Window

** The Embedding Layer and the Question of Context-Free Meaning*

The transformer processes a sequence of tokens within a bounded window called the context window. At the input stage (layer 0), each token is mapped to a vector via an embedding lookup table. At this

stage, the representation is context-independent: the embedding for "bank" is the same in every sentence.

This fact requires careful handling. The transformer does have a layer at which representations are context-free. However, these initial embeddings are not what drives the model's predictions. After the first layer of self-attention, every token's representation becomes a weighted function of every other token's representation in the context window. The self-attention operation computes:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$$

where Q, K, and V are the query, key, and value matrices and d_k is the key dimension (Vaswani et al., 2017).

A complication arises from residual connections. In a standard transformer, the output of each attention layer is added to its input via a residual pathway. This means that some signal from the original embedding can, in principle, persist through the network. Interpretability work has shown that token identity and other input-like features remain linearly decodable deep into the network via this residual stream structure (Nostalgebraist, 2020; Elhage et al., 2021). The context-free embedding is not erased; components of it remain linearly decodable in upper layers.

What this means for the present analysis is that the claim about transformers must be stated carefully. The claim is not that context-free information is entirely absent from upper-layer representations. It is that the transformer architecture does not implement an explicit sequential pipeline in which context-free meanings are first computed and then composed. The residual stream may carry forward fragments of the original embedding, but the architecture provides no mechanism that treats those fragments as independently meaningful units to be assembled. The distinction is between architectural availability (the signal persists in the residual stream) and functional role (the model does not use that signal as an independent meaning to be composed). Whether residual fragments of the embedding play any role in downstream prediction is an empirical question. Study 4 below provides relevant evidence.

**** Multi-Head Attention***

Transformers run multiple attention operations in parallel (multi-head attention), each with independently learned Q, K, and V projections. Different heads specialize in different types of relationships. Clark et al. (2019) and Voita et al. (2019) showed that some heads attend primarily to adjacent tokens, others to syntactically related tokens; more recent work has identified heads that track

semantic relations between tokens, including knowledge-graph-linked entities (Ren et al., 2024). Elhage et al. (2021) and Olsson et al. (2022) demonstrated that heads interact across layers to form circuits implementing complex sequential behaviors, such as in-context pattern completion.

The consequence is that the relationship between any two tokens is not a single scalar value but a multi-dimensional structure. A given token may be strongly related to another along one dimension (e.g., syntactic dependency) and weakly related along another (e.g., semantic similarity). This distributed, multi-dimensional character will be relevant when considering whether the Mīmāṃsā triad can map onto separable dimensions of the attention mechanism.

**** Positional Encoding***

Self-attention in its basic form is permutation-invariant: it computes pairwise relationships without regard to token order. Positional information must be added explicitly. Vaswani et al. (2017) used sinusoidal encodings. More recent architectures use relative position schemes. Su et al. (2024) introduced Rotary Position Embedding (RoPE), in which relative position is encoded through rotation of query and key vectors. Press et al. (2022) proposed ALiBi, which adds a learned linear bias to attention scores based on token distance.

In implementations with relative bias terms, tokens closer together tend to attend to each other more strongly. This is not a universal property of all transformer variants: vanilla attention with absolute encodings does not enforce monotonic decay. However, Clark et al. (2019) and Kovaleva et al. (2019) documented that in trained BERT models, attention mass frequently concentrates on a small number of tokens, including adjacent tokens and special delimiter tokens, even without explicit positional bias.

5. The Proposed Mapping

This section proposes that the three Mīmāṃsā conditions can be mapped onto three distinguishable aspects of the attention mechanism. It is important to state clearly what kind of claim this is. The mapping is structural: it identifies a correspondence between the functional roles of components in two different frameworks. Whether this structural correspondence carries explanatory weight, or whether it is merely a re-description, depends on whether it generates predictions or distinctions that the frameworks could not generate independently. Section 6 provides those tests.

**** Ākāṅkṣā and Query-Key Matching***

Ākāṅkṣā describes mutual expectancy: words syntactically and semantically "reach for" one another. A verb expects a subject. An accusative noun expects a transitive verb. This condition is mutual and pairwise: each word has expectations that specific other words fulfill.

The query-key mechanism of self-attention has a similar structure. Each token's query vector encodes what that token "seeks" from other tokens. Each token's key vector encodes what it "offers" to tokens that might attend to it. The dot product of Q and K determines the degree of mutual relevance. Like *ākāṅkṣā*, the mechanism is pairwise and, in bidirectional models, mutual (though not symmetric in magnitude). Clark et al. (2019) showed that certain attention heads specialize in attending from a word to its syntactic dependent or governor, which is precisely the kind of directed expectancy that *ākāṅkṣā* describes.

The correspondence is not identity. *Ākāṅkṣā* is a philosophical concept within a theory of linguistic cognition; query-key matching is a differentiable numerical operation. But the structural parallel is precise enough to generate testable predictions: if the mapping is correct, attention heads should systematically track obligatory syntactic dependencies more than optional ones, and the Prābhākara and Bhāṭṭa positions should generate different predictions about what happens when those tracking pathways are disrupted.

**** Yogyatā and Semantic Compatibility Constraints***

Yogyatā is the requirement that combined meanings be semantically compatible. The sentence *"irrigate with fire"* fails *yogyatā* because fire and irrigation are semantically incompatible. The failure is not syntactic (the sentence is grammatical) but semantic, determined by knowledge of the world.

In the transformer, semantic compatibility constraints are expressed in multiple ways: in attention patterns (the model attends less to semantically incompatible tokens), in MLP layers (which can suppress or promote semantic features), and in output logits (the model assigns low probability to incoherent continuations). Causal tracing methods, developed by Meng et al. (2022) to localize factual recall in GPT-style models, can in principle be adapted to identify which components are responsible for suppressing a specific incoherent continuation. Wang et al. (2023) showed that by substituting activations between paired inputs that differ only in a single target property, one can identify which attention heads and MLP layers are causally responsible for the output difference. If a controlled minimal pair is constructed (e.g., "irrigate the field with [water/fire]"), it should be possible to trace

which attention heads or MLP features respond differently to the compatible versus incompatible case. This would localize *yogyatā* to specific computational mechanisms rather than treating it as a diffuse property of the whole model. Section 6.2 reports exactly such experiments.

**** Sannidhi and the Bounded Context Window***

Sannidhi requires temporal or textual proximity. Words too far apart in discourse cannot form a sentence. The context window enforces precisely this constraint: tokens outside the window are invisible to the model. Within the window, in architectures that use relative position bias, proximity increases attention strength. The correspondence is straightforward. *Sannidhi* sets a boundary on the scope within which meaning-constitutive relations can operate. The context window sets a boundary on the scope within which attention can operate. Both acknowledge that relational meaning requires proximity.

**** Summary and Predictions***

Table 1 summarizes the proposed correspondence and the empirical tests reported in Section 6.

Condition	Definition	Transformer Analogue	Tested Via	Study
Ākāṅkṣā	Mutual expectancy	Query-key matching; syntactic heads	Attention + arc knockout	Study 1
Yogyatā	Semantic fitness	MLP layers; logit suppression	Activation patching	Study 2
Sannidhi	Contiguity / proximity	Context window; position bias	Decay curves; fact retrieval	Study 3

Table 1. Proposed mapping between Mīmāṃsā sentential conditions and transformer components, with empirical tests.

6. Experimental Evidence

All experiments were conducted on an NVIDIA A100-SXM4-80GB GPU using the TransformerLens library (Nanda et al., 2022) for GPT-2 and LLaMA, and HuggingFace Transformers for BERT. Models tested: BERT-base-uncased (12 layers, 12 heads), GPT-2 (12 layers, 12 heads), and LLaMA 3.1 8B (32 layers, 32 heads). Full code and datasets are available as a supplementary repository.

6.1 Study 1: Ākāṅkṣā and Syntactic Expectancy in Attention

**** Data and Method***

500 sentences (5 to 30 words) from the Universal Dependencies English Web Treebank (UD EWT), with gold-standard dependency parses. Dependency arcs were classified as obligatory (*ākāṅkṣā*-type:

nsubj, obj, iobj, csubj, ccomp, xcomp, cop, aux, mark, expl, nsubj:pass) or optional (advmod, amod, nmod, obl, advcl, appos, nummod, acl, acl:relcl). Word-to-token alignment was performed via BertTokenizerFast.word_ids(). For each of BERT's 144 attention heads, mean attention weight was computed separately along obligatory and optional arcs (dep-to-head and head-to-dep), aggregated at the sentence level. A paired t-test was performed for each head with Benjamini-Hochberg FDR correction across all 144 heads. Distance-binned analysis controlled for the arc length confound.

A causal test completed the analysis. For each obligatory arc, a distance-matched optional arc from the same sentence was identified. Attention along each arc was blocked via a custom 4D attention mask passed to the BERT encoder. Log-probability damage to the dependent token was measured under masked language modeling. The paired design controls for both distance and sentence-level variation.

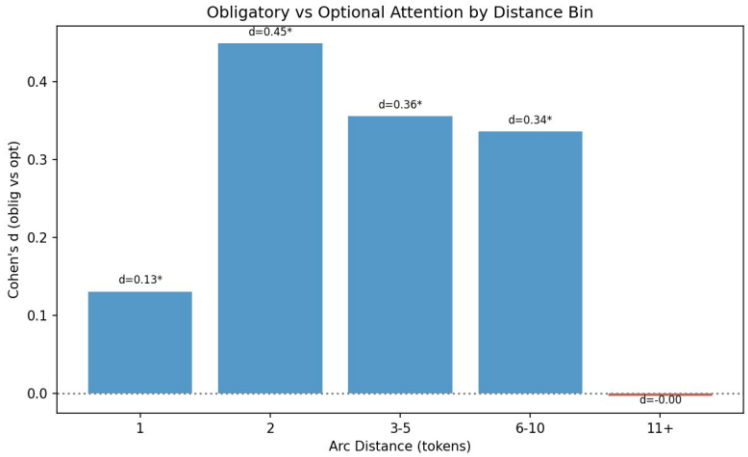

Figure 1. Obligatory vs. optional attention by arc distance. The effect is strongest at distance 2 tokens and persists up to 10 tokens before vanishing at long range.

** Results*

Token alignment was 100% (4,362 of 4,362 arcs). Of 144 attention heads, 94 (65.3%) showed significantly higher attention to obligatory than optional arcs (FDR < 0.05, bidirectional average). The top heads showed substantial effect sizes: layer 4 head 10 (Cohen's d = 0.80), layer 7 head 1 (d = 0.75), layer 6 head 9 (d = 0.66). The effect concentrated in middle layers (4 through 8), consistent with prior findings that syntactic processing peaks in mid-layers (Clark et al., 2019). Distance-binned analysis showed the obligatory-vs-optional attention difference was robust from 1 to 10 tokens (all d > 0.13, all p < 0.05), collapsing only beyond distance 11. (Figure 1 & 2).

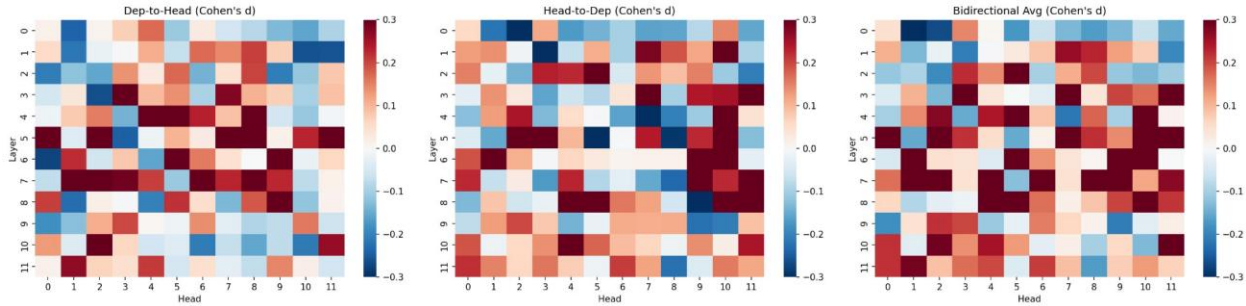

Figure 2. Effect size (Cohen's d) for obligatory vs. optional dependency attention across BERT layers and heads. Left: dependent-to-head direction. Center: head-to-dependent. Right: bidirectional average.

The paired arc knockout yielded a result that initially appears counterintuitive but is, on reflection, the more philosophically interesting finding. Across 484 distance-matched pairs, obligatory arcs caused *less* damage when knocked out than optional arcs (mean damage: obligatory = 1.90, optional = 2.74; mean difference = -0.83; paired t = -4.28, $p = 2.23 \times 10^{-5}$; Cohen's d = -0.20; Wilcoxon W = 47,357, $p = 2.34 \times 10^{-4}$). The effect was strongest at short distances (1 to 2 tokens) and diminished at longer ranges (6 to 10). (Figure 3).

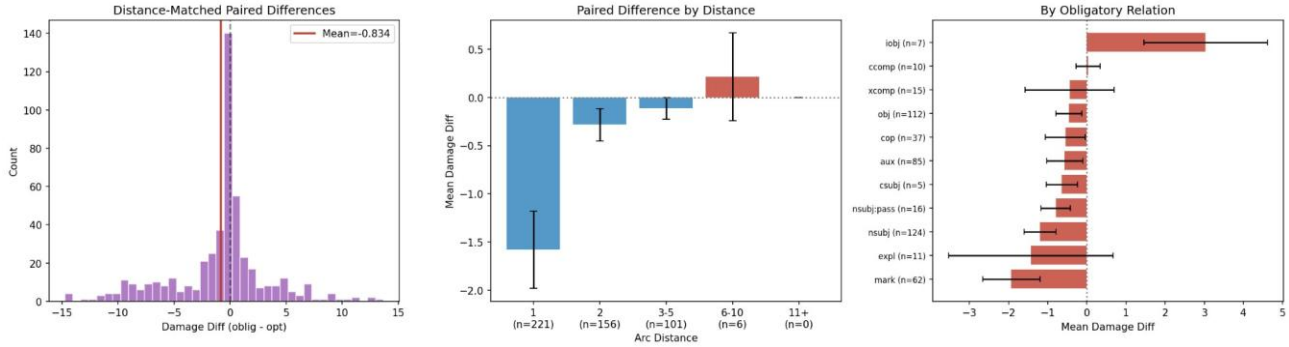

Figure 3. Paired arc knockout. Left: distribution of damage differences (obligatory minus optional). Center: by arc distance. Right: by obligatory relation type. Negative values indicate obligatory arcs are more robust to knockout.

** Constitutive vs. Supplementary Encoding*

The observational result (94 heads tracking obligatory arcs) and the causal result (knockout causing less damage to obligatory arcs) together form an argument with the following structure.

Premise 1: If *ākāṅkṣā*-type relations are supplementary (added to independently meaningful representations, the Bhāṭṭa view), then disrupting the attention pathway along a single obligatory arc should cause significant damage, because that pathway carries the combinatory signal.

Premise 2: If *ākāṅkṣā*-type relations are constitutive (built into the representation from the start, the Prābhākara view), then the information should be redundantly encoded across many pathways, and disrupting any single arc should cause relatively less damage.

Premise 3: The experiment shows that obligatory arc knockout causes less damage than optional arc knockout at matched distances.

Conclusion: The computational evidence is more consistent with the constitutive picture than the supplementary picture.

Why does this make sense? Obligatory dependencies (nsubj, obj, aux) encode highly predictable, redundant syntactic relationships. BERT can reconstruct a masked subject from the verb alone, from agreement features, from word order. The information flows through many paths; the heatmaps show exactly this, with many heads tracking the same obligatory arcs. Optional dependencies (adverbials, modifiers, appositives) carry less predictable, more idiosyncratic information. When the one attention path carrying a modifier's signal is disrupted, the damage is larger because there is no redundant path.

The mutual expectancy encoded by obligatory relations is so deeply distributed across the model's representational structure that no single arc knockout can substantially disrupt it. This distributed redundancy is itself evidence for the Prābhākara claim: these relations are constitutive of the representation, woven into its fabric, not supplementary signals added to pre-existing independent meanings. An analogy: consider the structural members of a building versus its decorative moldings. Removing one structural beam may cause little visible damage if the load is distributed across a redundant frame. Removing a unique decorative element produces an immediately visible gap. The lower damage from obligatory knockout does not mean these relations are unimportant. It means they are so important that the model has encoded them everywhere.

6.2 Study 2: Yogyatā and the Localization of Semantic Fitness

**** Data and Method***

200 minimal pairs (199 usable), 40 per violation type (agent, instrument, location, patient, temporal), drawn from a curated dataset of selectional restriction violations. Each pair shares a prompt prefix and diverges into plausible vs. implausible completion. Three analyses: (a) baseline behavioral discrimination via normalized logit difference (NLD) on LLaMA 3.1 8B; (b) noise-corruption activation patching, corrupting the full embedding with Gaussian noise ($\sigma = 3\times$ embedding std) and restoring clean activations one component at a time per layer; (c) semantic swap activation patching, using the good sentence as clean run and bad sentence as corrupt run, patching attention or MLP output from clean into corrupt at the final token position only. Both LLaMA and GPT-2 tested.

**** Results***

The model reliably distinguished fit from unfit completions. Mean NLD = 9.35 (Cohen's d = 2.67, p = 8.42×10^{-93}). 199 of 200 pairs classified correctly (99.5%). All five violation types were well-discriminated, with temporal violations showing the highest NLD (11.28) and instrument violations the lowest (6.56). (Figure 4).

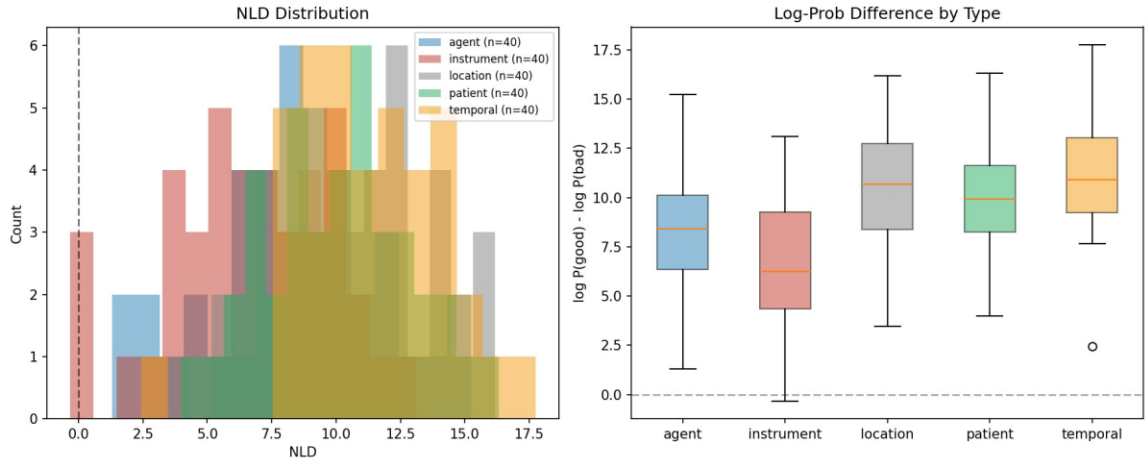

Figure 4. Yogyata behavioral results. Left: NLD distribution by violation type, nearly all positive. Right: log-probability difference (log P(good) minus log P(bad)) by violation type.

Noise patching revealed progressive buildup of *yogyatā* across layers. The residual stream showed smooth cumulative recovery peaking at layer 30 (recovery = 1.11). At the component level, MLP layers dominated: MLP output peaked at layer 31 (recovery = 0.38), while attention peaked at layer 21 (recovery = 0.14). Semantic swap patching confirmed the pattern: in LLaMA, MLP peaked at the last layer (0.24), attention at layer 26 (0.09). In GPT-2, attention dominated slightly (peak layer 11, recovery = 0.22) over MLP (layer 11, recovery = 0.09), suggesting the attention/MLP division of labor is architecture-dependent while the overall late-layer concentration is robust. (Figure 5).

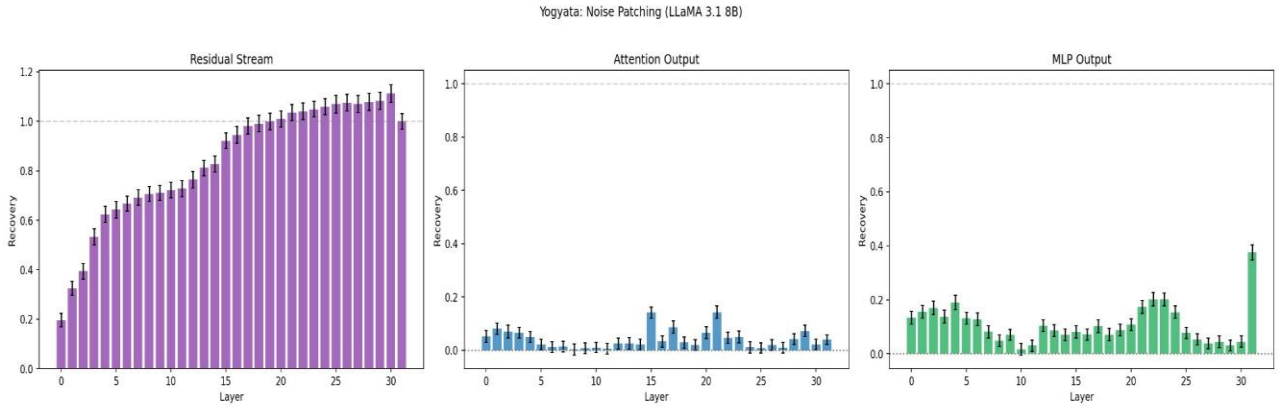

Figure 5. Activation patching localization of yogyata in LLaMA 3.1 8B. Left: residual stream shows progressive cumulative recovery. Center: attention output contribution is sparse and concentrated in layers 14-20. Right: MLP output shows distributed contribution across layers.

Appendix Figure A1 shows semantic swap patching results for both models. In LLaMA, MLP contribution is concentrated in the final layer (peak layer 31, recovery = 0.236), with attention contributing at layer 26 (recovery = 0.089). In GPT-2, the attention/MLP balance shifts: attention peaks at layer 11 (recovery = 0.220) and clearly exceeds MLP (peak layer 11, recovery = 0.086), indicating that late-layer concentration is robust across architectures while the division of labor between attention and MLP is architecture-dependent.

**** The Distributed Nature of Yogyatā***

The Mīmāṃsā tradition treats *yogyatā* as a single binary judgment: a sentence either satisfies semantic fitness or it does not. The *vahninā siñcati* test has a yes-or-no answer. But the computational evidence reveals something finer. Semantic fitness is computed gradually across layers, with different components contributing different amounts. There is no "*yogyatā* module." Instead, the ability to distinguish "the surgeon examined the wound with a scalpel" from "...with a cloud" emerges from the cumulative contributions of many layers, with MLP layers doing most of the heavy lifting.

This is a case where computational evidence refines the philosophical concept. The Mīmāṃsā taxonomy correctly identifies the *type* of constraint: *yogyatā* is genuinely distinct from *ākāṅkṣā* (the processing profiles are different; syntactic expectancy peaks in mid-layers while semantic fitness peaks in late layers) and from *sannidhi* (which operates at the level of attention weights over distance, not component-level logit recovery). But the implementation is more fine-grained than the classical theory anticipated. What the tradition called a judgment is, computationally, a gradient.

A further point concerns the relation between *yogyatā* and world knowledge. The Mīmāṃsā tradition recognized that semantic fitness depends on knowledge of the world, not just knowledge of language. Fire cannot irrigate because of what fire and irrigation are, not because of grammar. The dominance of MLP layers in the *yogyatā* computation is consistent with this: interpretability research has shown that MLP layers store factual and relational knowledge (Meng et al., 2022; Geva et al., 2021), while attention layers route information (Elhage et al., 2021). That the transformer localizes semantic fitness primarily in its knowledge-storing components is a structural alignment with the classical insight that *yogyatā* draws on world knowledge.

Does this mean the model "understands" semantic fitness? When LLaMA assigns low probability to "she cut the rope with a cloud," it is not exercising genuine semantic understanding. It is exploiting

statistical regularities. But the Mīmāṃsā concept of *yogyatā* does not require understanding in the phenomenological sense. It requires that the system reliably distinguish fit from unfit combinations. The transformer satisfies the functional criterion of *yogyatā* without satisfying the phenomenological criterion of understanding. This distinction is itself productive: it separates the structural condition from the experiential condition that the classical tradition may have assumed were inseparable.

6.3 Study 3: Sannidhi and the Effective Context Radius

**** Data and Method***

313 prose passages (130 to 200 tokens) for short-range analysis; 50 long passages (~1,000 tokens) for long-range analysis. Three experiments: (a) *Attention decay*: mean attention weight vs. token distance, fit to exponential $A \cdot \exp(-d/\tau)$; (b) *Long-range decay* on LLaMA (RoPE), 1,000-token passages; (c) *Functional distance*: entity log-probability after passage truncated to various distances (50, 100, 200, 400, 600, 800 tokens), followed by a retrieval query.

**** Results***

Attention decay followed an exponential profile in all three models. The fitted time constants (τ): BERT $\tau = 2.65$, GPT-2 $\tau = 3.50$, LLaMA $\tau = 5.69$. These define model-specific contiguity radii. LLaMA's larger τ is consistent with its use of RoPE. Layer-by-layer analysis showed earlier layers attend more locally (steeper decay), while later layers show flatter profiles. (Figure 6).

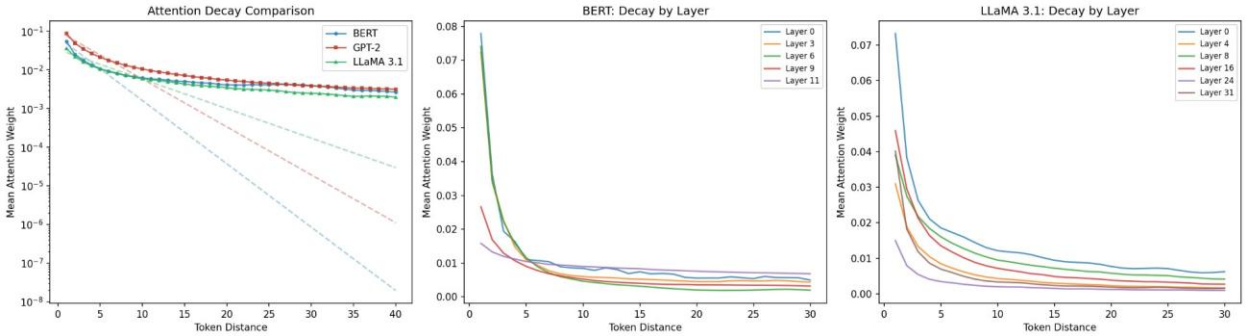

Figure 6. Sannidhi: attention decay as a function of token distance. Left: three-model comparison with exponential fits (dashed). Center: BERT decay by layer. Right: LLaMA 3.1 decay by layer.

Long-range decay in LLaMA showed exponential decay in the first 50 to 100 tokens ($\tau = 14.6$), then a plateau at $\sim 10^{-4}$ rather than continuing to decay. Layer 0 showed the highest long-range attention, with deeper layers concentrating more on local context (Figure A2).

Functional distance testing confirmed retrieval degrades with distance. Mean regression slope of log P(entity) on distance was -0.00094, below zero ($t = -4.15$, $p = 1.31 \times 10^{-4}$, Cohen's $d = -0.59$). The effect was steepest in the first 200 tokens and then leveled off. (Figure 7).

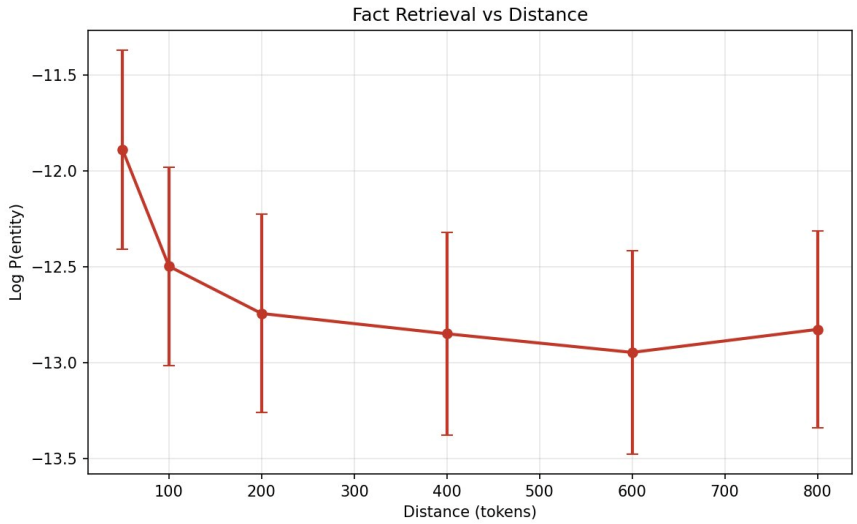

Figure 7. Functional sannidhi: log-probability of a passage-initial entity decays as the query is moved farther from the fact. Error bars show standard error across passages.

* Sannidhi as Transcendental Condition

Sannidhi is the most interesting condition because it is not merely an empirical regularity but a condition of possibility. In the Mīmāṃsā framework, *sannidhi* is not the observation that words closer together tend to be understood together. It is the claim that proximity is a *necessary condition* for sentential meaning to arise at all. Words spoken on different days cannot form a sentence, regardless of their *ākāṅkṣā* and *yogyatā* relations.

The context window is the architectural equivalent of this transcendental claim. It is not a soft preference but a hard boundary: tokens outside the window literally cannot participate in the computation. The attention decay data shows the soft gradient within the window. But the window boundary itself is the more significant feature. It means that the transformer, like the Mīmāṃsā theory, treats proximity as a precondition for meaning, not merely a factor that influences it.

The Mīmāṃsā tradition debated what counts as "sufficient proximity." Is hearing a sentence with a long pause in the middle one sentence or two? The functional distance experiment is an empirical version of this debate. The exponential decay curve with τ values gives a quantitative *sannidhi* radius for each model: a philosophical concept (the boundary of sentential unity) given a precise empirical

measurement. BERT's radius ($\tau = 2.65$) is narrow; LLaMA's ($\tau = 5.69$ short-range, 14.6 long-range) is substantially wider. The classical tradition had no way to quantify this boundary. The computational evidence provides one.

The plateau in LLaMA's long-range decay is also philosophically suggestive. Even at 500 tokens, attention does not drop to zero; it stabilizes at a floor. The *sannidhi* condition does not simply switch off at a threshold; it degrades asymptotically. The classical formulation "words spoken on different days" marks the extreme case, but the computational evidence suggests that *sannidhi* is better understood as a graded condition with a steep initial decline and a long tail.

6.4 Study 4: Polysemy as Decision Point

Polysemy provides a useful test case because it is a situation where the question "does a context-free word meaning exist as a computational object?" becomes empirically tractable. Consider "bank" in two sentences: (a) "She walked along the river bank" and (b) "She walked into the bank to deposit a check."

On the *abhihitānvaya* account, "bank" has a stable polysemous denotation (something like BANK, encompassing all senses) which is then disambiguated by sentential context. The word first means BANK; then context selects the correct sense. On the *anvitābhidhāna* account, there is no stage at which "bank" means BANK in an undifferentiated way. In sentence (a), "bank" denotes bank-as-geographical-feature-of-river; in sentence (b), it denotes bank-as-financial-institution-entered-for-deposit. The sense is constituted by the relational context, not selected from a pre-existing menu.

Standard accounts of polysemy resolution in NLP describe the process as "contextual modulation of a base representation." The Prābhākara framework suggests a different description: not modulation of a pre-existing meaning, but constitution of meaning through relational context. Whether "modulation" or "constitution" is the better description is an empirical question. The two theories make incompatible predictions about what should be observed across layers.

** Data and Method*

1,950 sentences covering 39 polysemous English words, each with two established senses and 25 sentences per sense (50 per word). Representations were extracted from BERT-base-uncased (13 layers including the embedding layer). BERT was selected because its bidirectional attention allows each target word representation to incorporate the full sentential context, which is necessary for clean

sense disambiguation; autoregressive models encode only left context at the target position, producing less sharp sense distinctions. Two metrics were computed at each layer: (a) sense silhouette, measuring how well representations of the same word cluster by contextual sense, and (b) identity silhouette, measuring how well representations cluster by lexical identity across all words. Crossover layer: the first layer at which sense separability exceeds identity separability. Bootstrap (500 resamples over words) with corrected identity labels for duplicate words.

** Results*

In BERT, lexical identity dominated strongly at layer 0 (silhouette = 0.79) and declined monotonically. Sense separability started near zero (0.03 at layer 0) and increased to 0.34 by layer 12. The crossover occurred at layer 6 (Figure 8).

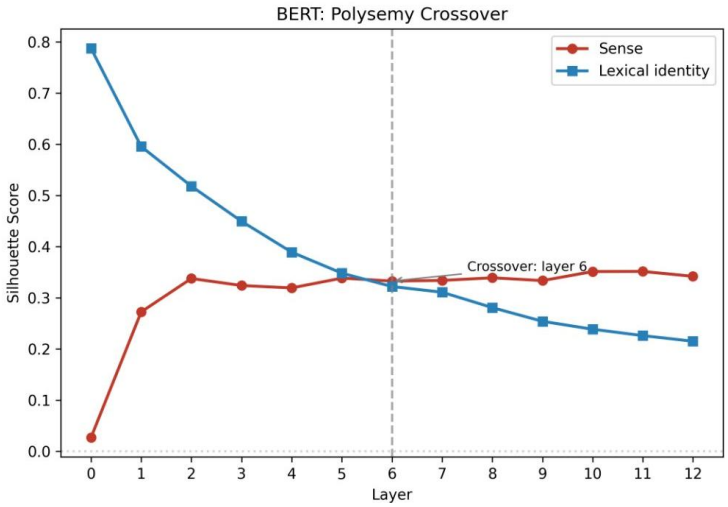

Figure 8. Polysemy crossover in BERT. Lexical identity silhouette (blue) declines monotonically; sense silhouette (red) increases. The crossover at layer 6 marks where representations are organized more by contextual meaning than surface form.

The bootstrap confirmed this robustly: 500 of 500 resamples showed a crossover, median at layer 7, 95% CI [5, 8]. At the final layer, sense (0.34) exceeded identity (0.22). (Figure 8; bootstrap distribution in Figure A3).

** Disconfirming a Prediction of Abhihitānvaya*

On the *abhihitānvaya* account, there should be a representational stage at which "bank" means BANK-in-general before context selects a sense. This predicts that lexical identity should remain the dominant organizing principle at every layer, with sense information gradually added on top.

On the *anvitābhidhāna* account, there should be no such stage in the model's functional processing. Sense-specific representations should emerge as soon as context begins to influence the representation, and should eventually dominate over lexical identity.

The BERT crossover at layer 6 is not merely "consistent with" *anvitābhidhāna*. It actively disconfirms a specific prediction of *abhihitānvaya*. The Bhāṭṭa theory predicts identity dominance throughout. That prediction fails. By layer 6, representations are organized by contextual sense, not lexical identity. The context-free embedding is not what the model uses for its actual semantic work. The residual stream carries fragments of it forward, but carrying forward and functionally relying on are different things. An analogy: a palimpsest carries traces of its original text, but no reader is reading the original. The traces persist physically without persisting functionally.

Prior literature supports these findings. Ethayarajh (2019) measured the cosine similarity of the same word's representation across different contexts at each layer and found representations become increasingly context-specific in upper layers. Reif et al. (2019) showed evidence that BERT's representations of polysemous words form geometrically distinct clusters by sense.. Wiedemann et al. (2019) confirmed that contextualized embeddings outperform static embeddings for word sense disambiguation. The present study adds the crossover metric, which makes the theoretical stakes precise: there is a specific layer at which the model's representational priorities shift from "what word is this?" to "what does this word mean here?"

7. Against the Embedding Layer Objection

The strongest argument available to *abhihitānvaya* is the existence of the embedding layer. At layer 0, the transformer has context-free representations. Does this vindicate the Bhāṭṭa picture?

The Bhāṭṭa claim is not merely that context-free representations exist at some stage. The claim is that those context-free meanings are functionally operative as independently meaningful units that are then composed (Matilal & Sen, 1988). For the embedding layer to vindicate *abhihitānvaya*, two conditions would need to hold.

First condition: the context-free embeddings would need to be the representations that drive downstream prediction. The polysemy crossover data (Study 4) directly addresses this. By layer 6 in

BERT, representations are organized by contextual sense, not lexical identity. The context-free embedding is not what the model uses for its actual semantic work.

Second condition: there would need to be an identifiable computational step that takes those independent meanings and combines them. This condition fails architecturally. There is no "compose" step in the transformer. There is no module that takes the output of an "independent meaning" stage and feeds it into a "combination" stage. The self-attention mechanism operates on all tokens simultaneously from the first layer onward. This is not an empirical finding but an architectural fact, and it maps precisely onto the Prābhākara critique of procedural compositionality.

Even granting the embedding layer, then, *abhihitānvaya* requires two things the transformer does not provide: context-free representations that are functionally operative, and an identifiable composition stage. The residual stream complicates the picture, since it carries forward fragments of the embedding. But the distinction between architectural availability and functional role is what matters. A palimpsest's original text is physically present but not what anyone reads. The embedding layer is more fairly characterized as an engineering constraint (a practical method for converting discrete symbols to continuous vectors) than as evidence for a stage of independent denotation.

8. Bhartṛhari's Sphoṭa as a Limiting Case

Bhartṛhari (c. 5th century CE) held, in the *Vākyapadīya*, that the sentence is the only real unit of meaning. Words are analytical abstractions from an indivisible whole. The meaning-bearing entity (*sphoṭa*) is revealed through the temporal sequence of phonemes but is not constructed from them (Houben, 1995; Iyer, 1969). This is a more radical position than *anvitābhidhāna*, which still grants words a relational denotative function.

If *anvitābhidhāna* corresponds to the transformer's relational processing of discrete tokens within a context window, *sphoṭa* would correspond to a hypothetical architecture in which there are no discrete tokens at all. Some recent work on state-space models (Gu et al., 2022) and continuous-space diffusion models (Li et al., 2022) reduces reliance on autoregressive token-by-token processing, operating instead over continuous or recurrent sequence representations. Whether these architectures will prove effective, and whether *sphoṭa* is the right framework for understanding them, remain open questions.

For the present argument, *sphoṭa* marks the boundary. The transformer retains discrete tokenization; it therefore retains a commitment to word-level units, even if it processes them relationally. Bhartṛhari would regard even *anvitābhidhāna* as insufficiently holistic, because it still speaks of words denoting, albeit relationally. The *sphoṭa* framework thus identifies what the transformer, with its discrete tokenization, cannot model, and thereby clarifies where the correspondence with the Mīmāṃsā tradition ends.

9. Objections and Limitations

**** Is This More Than Analogy?***

The most direct objection is that the mapping merely re-labels transformer components with Sanskrit terms. The response is that the mapping generates specific, testable predictions that the Sanskrit framework alone and the interpretability framework alone would not have generated independently. The constitutive-vs-supplementary argument (Study 1) follows from explicitly Prābhākara premises. The polysemy crossover test (Study 4) follows from the logical structure of the *anvitābhidhāna/abhihitānvaya* debate. These are not predictions that standard interpretability research would have formulated in the same terms. A bare analogy does not generate testable predictions. A structural mapping that specifies where to look and what to measure, and that produces confirmed results, earns more than terminological status.

**** The Models Are Not Minds***

This is correct and important. The Prābhākaras were making claims about the phenomenology of human linguistic cognition. Transformers have no phenomenology. The argument is not that transformers validate the Prābhākara position as a theory of mind. It is that the structural properties the Prābhākaras identified (constitutive contextual embedding, distributed encoding of obligatory relations, graded semantic fitness, proximity as a precondition for meaning) are realized in a system that processes language successfully. This is evidence that these properties are not merely features of human cognition but may be features of successful language processing as such. The philosophical interest lies in the convergence: two systems developed for entirely different purposes arrive at the same structural answer.

**** Methodological Caution in Cross-Traditional Comparison***

A reviewer in Indian philosophy may reasonably worry about anachronism. The Prābhākaras were working within a tradition of Vedic exegesis, not computational theory. Their concerns were hermeneutic and, in a broad sense, soteriological. The methodological principle followed here is the one articulated by Matilal (1986) and Ganeri (2001): the logical content of a philosophical argument can be evaluated independently of its motivating context. The logical content of *anvitābhidhāna* is a claim about the structure of denotation. This claim can be placed in dialogue with a claim about the structure of contextual representation in neural networks. The dialogue is productive insofar as it sharpens both claims. It is not a claim that either tradition explains, validates, or supersedes the other.

**** Statistical Limitations***

Two results deserve caution. The Study 1 knockout effect size (d = -0.20) is small, though statistically robust across heads. The Study 3 Spearman correlation for fact retrieval (r = -0.089) is weak, likely because the distance-attention relationship is strongly nonlinear and concentrated in the initial 200-token range, making rank correlation an imperfect measure. Taken together, these figures suggest the mapping is stronger for some conditions than others: the ākāṅkṣā observational result and the polysemy crossover are the most robust; the sannidhi functional distance result is the most provisional. A finer-grained account of when the mapping holds and when it weakens is a direction for future work

10. Conclusion

The Mīmāṃsā debate between *anvitābhidhāna* and *abhihitānvaya* is a debate about procedural compositionality: whether meaning-constitution proceeds through a stage of context-independent representation. The transformer attention mechanism, which computes contextual representations without an explicit compose step, has a structural affinity with the Prābhākara answer. This affinity is not incidental. The Prābhākara argument targets precisely the procedural sense of compositionality, the claim that independently meaningful representations must exist before combination, and this is also the sense in which the transformer architecture departs from the classical computational pipeline. The coincidence is structural, not merely terminological.

The Mīmāṃsā triad of *ākāṅkṣā*, *yogyatā*, and *sannidhi* offers more than a re-labeling of transformer components. It offers a taxonomy with a pre-existing internal logic, one developed over seven

centuries of sustained debate, tested against competing theories, and refined through primary-source argumentation. That taxonomy distinguishes types of relational constraint that the interpretability literature currently treats under a single heading.

The experimental evidence supports four conclusions. First, obligatory syntactic dependencies (*ākāṅkṣā*) are tracked by a majority of BERT's attention heads and are encoded with distributed redundancy, consistent with the Prābhākara claim that these relations are constitutive rather than supplementary. Second, semantic fitness (*yogyatā*) is computed as a graded, distributed process concentrated in MLP layers, refining the classical binary conception while confirming the distinctness of the constraint type. Third, contiguity (*sannidhi*) operates both as a hard boundary (the context window) and as a soft exponential gradient (attention decay), with model-specific quantitative radii that give the philosophical concept an empirical metric. Fourth, polysemy representations cross from identity-dominated to sense-dominated organization at a specific, statistically robust layer in BERT, disconfirming a prediction of *abhihitānvaya* and supporting the *anvitābhidhāna* account of meaning as relationally constituted.

What this paper does not claim is that the Mīmāṃsā framework explains transformer processing, or that transformer evidence validates Prābhākara philosophy. The two frameworks were developed for different purposes, under different evidentiary standards, in response to different problems. What the comparison does claim, and what the argument has tried to establish, is that placing them in dialogue produces distinctions precise enough to generate testable predictions, and a vocabulary exact enough to be useful to researchers in both traditions.

Future work might apply causal tracing to localize yogyata to specific circuits; examine whether the Nyāya school's distinct approach to sentence meaning maps onto different architectural choices; or investigate whether Bhartrhari's sphota framework sheds light on continuous-representation architectures. The Mimamsa tradition spent centuries developing the conceptual tools to ask precise questions about how language means. It is worth finding out how far those tools travel.

References

Arai, Y., & Tsugawa, S. (2024). Do large language models defend inferentialist semantics? On the logical expressivism and anti-representationalism of LLMs. https://arxiv.org/abs/2412.14501

Clark, K., Khandelwal, U., Levy, O., & Manning, C. D. (2019). What does BERT look at? An analysis of BERT's attention. Proceedings of the ACL Workshop BlackboxNLP, 276-286.

Csordás, R., Irie, K., & Schmidhuber, J. (2021). The devil is in the detail: Simple tricks improve systematic generalization of transformers. Proceedings of EMNLP, 619-634.

Deshpande, M. M. (1998). Meaning, Indian theories of. In E. Craig (Ed.), Routledge encyclopedia of philosophy. Taylor and Francis. https://doi.org/10.4324/9780415249126-F052-1

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., & Olah, C. (2021). A mathematical framework for transformer circuits. *Transformer Circuits Thread*. https://transformer-circuits.pub/2021/framework/index.html

Ethayarajh, K. (2019). How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 representations. Proceedings of EMNLP-IJCNLP, 55-65.

Firth, J. R. (1957). A synopsis of linguistic theory 1930-1955. In Studies in Linguistic Analysis (pp. 1-32). Blackwell.

Fodor, J. A., & Pylyshyn, Z. W. (1988). Connectionism and cognitive architecture: A critical analysis. Cognition, 28(1-2), 3-71.

Frege, G. (1892). Über Sinn und Bedeutung. Zeitschrift für Philosophie und philosophische Kritik, 100, 25-50.

Ganeri, J. (2001). Philosophy in Classical India: The Proper Work of Reason. Routledge.

Geva, M., Schuster, R., Berant, J., & Levy, O. (2021). Transformer feed-forward layers are key-value memories. Proceedings of EMNLP, 5484-5495.

Groenendijk, J., & Stokhof, M. (1991). Dynamic predicate logic. Linguistics and Philosophy, 14(1), 39-100. https://doi.org/10.1007/BF00628304

Gu, A., Goel, K., & Ré, C. (2022). Efficiently modeling long sequences with structured state spaces. Proceedings of ICLR.

Heim, I. (1982). The Semantics of Definite and Indefinite Noun Phrases (Doctoral dissertation, University of Massachusetts Amherst).

Hornby, R. (2025). Is generative AI ready to join the conversation that we are? Gadamer's hermeneutics after ChatGPT. Technophany, A Journal for Philosophy and Technology, 3(1), 1-31. https://doi.org/10.54195/technophany.18134

Houben, J. E. M. (1995). The Sambandha-Samuddesha and Bhartṛhari's Philosophy of Language. Egbert Forsten.

Iyer, K. A. S. (1969). Bhartṛhari: A Study of the Vākyapadīya in the Light of Ancient Commentaries. Deccan College.

Jha, G. (1942). Pūrva-Mīmāṃsā in Its Sources. Banaras Hindu University Press.

Kovaleva, O., Romanov, A., Rogers, A., & Rumshisky, A. (2019). Revealing the dark secrets of BERT. Proceedings of EMNLP-IJCNLP, 4365-4374.

Lake, B. M., & Baroni, M. (2018). Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. Proceedings of ICML, 2873-2882.

Lenci, A. (2008). Distributional semantics in linguistic and cognitive research. Italian Journal of Linguistics, 20(1), 1-31.

Li, X., Thickstun, J., Gulrajani, I., Liang, P., & Hashimoto, T. B. (2022). Diffusion-LM improves controllable text generation. Advances in NeurIPS, 35.

Matilal, B. K. (1986). Perception: An Essay on Classical Indian Theories of Knowledge. Oxford University Press.

Matilal, B. K. (1990). The Word and the World: India's Contribution to the Study of Language. Oxford University Press.

Matilal, B. K., & Sen, P. K. (1988). The context principle and some Indian controversies over meaning. *Mind*, *97*(385), 73–97. https://www.jstor.org/stable/2254913

McCrea, L. (2000). The hierarchical organization of language in Mīmāṃsā interpretive theory. Journal of Indian Philosophy, 28(5), 429-459. https://doi.org/10.1023/A:1017527826104

McCrea, L. (2020). Kumārila on the role of implicature in sentence-signification. In A. Graheli (Ed.), The Bloomsbury research handbook of Indian philosophy of language (pp. 353-360). Bloomsbury Publishing.

Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. Advances in NeurIPS, 35.

Montague, R. (1973). The proper treatment of quantification in ordinary English. In J. Hintikka et al. (Eds.), Approaches to Natural Language (pp. 221-242). Reidel.

Nanda, N., et al. (2022). TransformerLens. https://github.com/TransformerLensOrg/TransformerLens.

Nostalgebraist. (2020). *Interpreting GPT: The logit lens*. LessWrong. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru

Olsson, C., Elhage, N., Nanda, N., et al. (2022). In-context learning and induction heads. Anthropic Research.

Ontanon, S., Ainslie, J., Fisher, Z., & Cvicek, V. (2022). Making transformers solve compositional tasks. Proceedings of ACL, 3591-3607.

Patel, R., & Pavlick, E. (2022). Mapping language models to grounded conceptual spaces. Proceedings of ICLR.

Press, O., Smith, N. A., & Lewis, M. (2022). Train short, test long: Attention with linear biases enables input length generalization. Proceedings of ICLR.

Raja, K. K. (1963). Indian Theories of Meaning. Adyar Library and Research Centre.

Recanati, F. (2004). Literal Meaning. Cambridge University Press.

Reif, E., Yuan, A., Wattenberg, M., et al. (2019). Visualizing and measuring the geometry of BERT. Advances in NeurIPS, 32.

Ren, J., Guo, Q., Yan, H., Liu, D., Zhang, Q., Qiu, X., & Lin, D. (2024). Identifying semantic induction heads to understand in-context learning. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), Findings of the Association for Computational Linguistics: ACL 2024 (pp. 6916–6932). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-acl.412

Schwitzgebel, E. (2008). The unreliability of naive introspection. Philosophical Review, 117(2), 245-273.

Smolensky, P. (1990). Tensor product variable binding and the representation of symbolic structures in connectionist systems. Artificial Intelligence, 46(1-2), 159-216.

Smolensky, P., & Legendre, G. (2006). The Harmonic Mind: From Neural Computation to Optimality-Theoretic Grammar. MIT Press.

Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., & Liu, Y. (2024). RoFormer: Enhanced transformer with Rotary Position Embedding. Neurocomputing, 568, Article 127063. https://doi.org/10.1016/j.neucom.2023.127063

Szabó, Z. G. (2020). Compositionality. In E. N. Zalta (Ed.), The Stanford encyclopedia of philosophy (Fall 2020 ed.). Metaphysics Research Lab, Stanford University

Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. Advances in NeurIPS, 30, 5998-6008.

Voita, E., Talbot, D., Moiseev, F., Sennrich, R., & Titov, I. (2019). Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. Proceedings of the 57th ACL, 5797-5808.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., & Steinhardt, J. (2023). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *Proceedings of the International Conference on Learning Representations (ICLR 2023)*. https://arxiv.org/abs/2211.00593

Wiedemann, G., Remus, S., Chawla, A., & Biemann, C. (2019). Does BERT make any sense? Interpretable word sense disambiguation with contextualized embeddings. Proceedings of KONVENS.

Zhu, X., Zhou, Z., Charlow, S., & Frank, R. (2025). Meaning beyond truth conditions: Evaluating discourse level understanding via anaphora accessibility. https://arxiv.org/abs/2502.14119

Appendix: Supplementary Figures

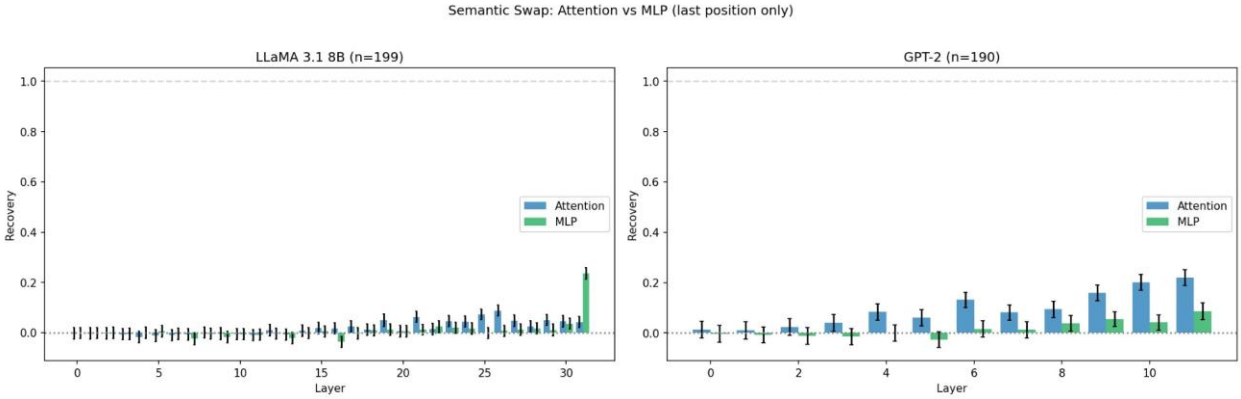

Figure A1. Semantic swap patching (attention vs. MLP) for LLaMA 3.1 8B and GPT-2. Both models show MLP contribution concentrated in final layers, with attention contributing primarily in upper layers.

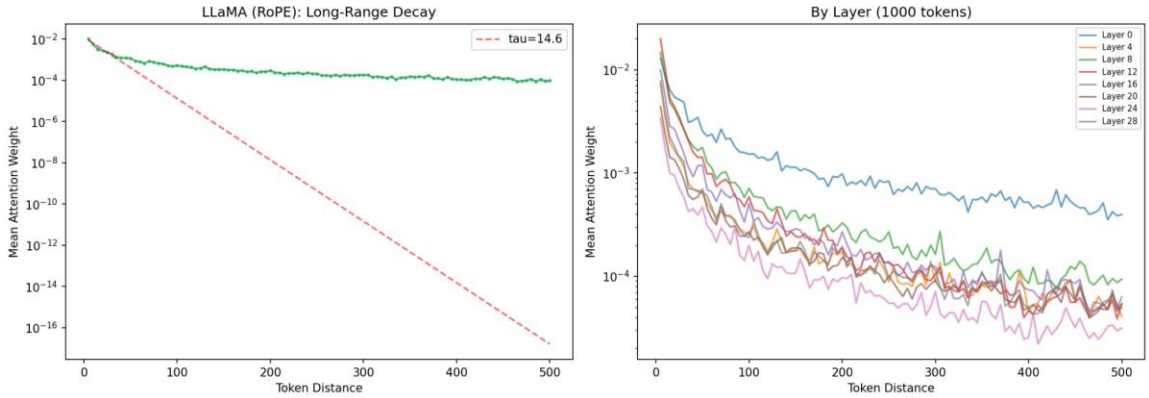

Figure A2. Long-range attention decay in LLaMA (RoPE) over 1000-token passages. Left: overall decay with exponential fit (tau = 14.6). The curve plateaus at ~10^-4 rather than continuing to decay. Right: by-layer profiles showing layer 0 retains the broadest attention window.

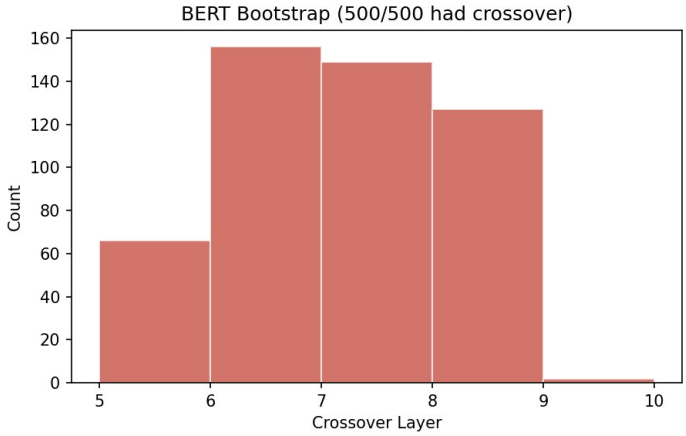

Figure A3. Bootstrap distribution of polysemy crossover layer in BERT (500 resamples). All 500 resamples showed a crossover, with median at layer 7, 95% CI [5, 8].